EXPERIMENT 35: GASTROINTESTINAL MOTILITY TESTS

40 STANDARD IN-COURSE MCQs

1. The primary aim of Experiment 35 is to determine:

A. The gastric acid output in anesthetized rats B. The effects of pharmacological agents on gastrointestinal motility and transit using the charcoal meal test C. The degree of mucosal ulceration following ethanol challenge D. The anti-inflammatory effect on rat paw volume

Answer: B

2. What experimental animals are designated in the protocol on Page 164?

A. Wistar rats B. Albino mice (either sex) C. Guinea pigs D. New Zealand rabbits

Answer: B

3. How many mice are allocated to each experimental group in this protocol?

A. 2 mice B. 5 mice C. 10 mice D. 20 mice

Answer: B

4. What fasting duration is specified for the mice prior to the test?

A. 30 minutes B. 2 to 8 hours C. 24 to 48 hours D. No fasting

Answer: B

5. Why is fasting necessary before performing an intestinal transit test?

A. To prevent mice from sleeping B. To ensure uniform gastrointestinal filling and eliminate variable physical obstruction by food boluses C. To destroy intestinal bacteria D. To deplete liver glycogen stores

Answer: B

6. What marker meal is administered to visualize intestinal transit?

A. Barium sulfate suspension B. Deactivated charcoal meal suspended in aqueous gum tragacanth or acacia C. Methylene blue dye in alcohol D. Radiopaque iodine beads

Answer: B

7. Why is activated/deactivated charcoal chosen as the intestinal marker?

A. It is actively absorbed into the bloodstream B. It is non-absorbable, physiologically inert, and easily visualized as a dense black front against the pink bowel C. It stimulates rapid smooth muscle contraction D. It destroys all intestinal enzymes

Answer: B

8. What is the role of aqueous gum tragacanth or acacia in the charcoal meal preparation?

A. It acts as an active laxative B. It acts as a suspending/viscosity agent to keep the dense charcoal particles evenly dispersed without settling C. It neutralizes hydrochloric acid D. It accelerates gastric emptying

Answer: B

9. Which agent is administered as the negative control in Group I?

A. Normal saline or distilled water (10 mL/kg p.o.) B. Olive oil C. Absolute alcohol D. Glucose solution

Answer: A

10. Which classical antimuscarinic drug is administered as the positive inhibitory control?

A. Neostigmine B. Atropine (0.1–1.0 mg/kg i.p. or s.c.) C. Pilocarpine D. Metoclopramide

Answer: B

11. How much time elapses between administering the test/control drugs and giving the oral charcoal meal?

A. 5 minutes B. 1 hour (or 30–60 minutes) C. 6 hours D. 24 hours

Answer: B

12. How much time is allowed for the charcoal meal to travel along the intestine before sacrificing the animal?

A. 5 minutes B. 30 minutes C. 3 hours D. 24 hours

Answer: B

13. Why is the animal sacrificed precisely 30 minutes after charcoal administration?

A. The animal dies naturally at 30 minutes B. In normal mice, 30 minutes allows the charcoal front to travel through a measurable portion of the small intestine without passing completely into the cecum C. Charcoal degrades after 30 minutes D. It is the time required for complete gastric emptying of all solids

Answer: B

14. How are the mice humanely sacrificed at the end of the 30-minute transit period?

A. Decapitation without anesthesia B. Cervical dislocation C. High-dose chloroform exposure D. Freezing

Answer: B

15. What surgical steps follow immediately after sacrificing the animal?

A. Opening the thorax to harvest the heart B. Laparotomy, carefully exposing and excising the small intestine from the pyloric sphincter to the ileocecal junction C. Excision of the colon and rectum only D. Harvesting the kidneys

Answer: B

16. What two precise lengths are measured along the excised small intestine?

A. Stomach diameter and cecal width B. Total length of the small intestine (cm) and distance travelled by the charcoal marker front from the pylorus (cm) C. Duodenal width and colon length D. Spleen length and liver circumference

Answer: B

17. What mathematical formula gives the percentage intestinal transit (% Motility)?

A. (Total length / Distance of charcoal) * 100 B. (Distance travelled by charcoal meal / Total length of small intestine) * 100 C. (Distance of charcoal * Total length) / 100 D. (Total length - Distance of charcoal) / 100

Answer: B

18. If a mouse has an intestinal length of 50.0 cm and the charcoal front moved 35.0 cm, what is the % transit?

A. 35% B. 50% C. 70% D. 85%

Answer: C

19. In a mouse pre-treated with atropine, what effect is expected on the distance travelled by the charcoal front?

A. The charcoal travels significantly farther, reaching 100% of the intestine B. The charcoal travels a much shorter distance, resulting in a marked decrease in % transit C. The charcoal completely disappears from the gut D. The charcoal moves backwards into the mouth

Answer: B

20. What is the molecular mechanism of atropine in reducing gastrointestinal transit?

A. Stimulation of nicotinic receptors in sympathetic ganglia B. Competitive blockade of muscarinic M_3 receptors on intestinal smooth muscle, inhibiting peristaltic contractions C. Direct inhibition of smooth muscle actin D. Activation of beta-2 adrenergic receptors

Answer: B

21. What effect would a prokinetic drug like metoclopramide produce on intestinal transit?

A. Complete cessation of intestinal movement B. Significant increase in charcoal transit distance and higher % motility C. Severe intussusception D. Discoloration of the intestinal wall to white

Answer: B

22. Metoclopramide accelerates upper gastrointestinal transit primarily by:

A. Blocking 5-HT_3 receptors and agonizing 5-HT_4 receptors while blocking dopamine D_2 receptors in the enteric plexus B. Blocking muscarinic M_3 receptors C. Activating opioid mu receptors D. Inhibiting cyclic AMP production

Answer: A

23. How do opioid analgesics (e.g., morphine, codeine, loperamide) affect intestinal motility?

A. Increase peristalsis and cause profuse diarrhea B. Activate intestinal mu-opioid receptors, decreasing peristaltic propulsive waves and increasing sphincter tone, causing severe constipation C. Selectively dilate the ileocecal valve D. Stimulate acetylcholine release from the myenteric plexus

Answer: B

24. Loperamide is widely used clinically as an antidiarrheal because:

A. It is a potent central analgesic B. It is a peripheral mu-opioid receptor agonist that does not cross the blood-brain barrier at therapeutic doses C. It is an antibiotic that kills E. coli D. It completely neutralizes gastric acid

Answer: B

25. Which division of the autonomic nervous system normally stimulates gastrointestinal motility and secretions?

A. Sympathetic nervous system (thoracolumbar outflow) B. Parasympathetic nervous system (craniosacral outflow via the vagus and pelvic splanchnic nerves) C. Somatosensory nervous system D. Corticospinal motor tract

Answer: B

26. Which intrinsic neural network in the gut wall directly coordinates gastrointestinal peristalsis?

A. Meissner's submucosal plexus B. Auerbach's myenteric plexus (located between circular and longitudinal muscle layers) C. Brachial plexus D. Celiac plexus

Answer: B

27. If an unknown plant extract significantly reduces charcoal transit from 65% to 25%, the extract possesses:

A. Prokinetic / laxative properties B. Antispasmodic / antidiarrheal properties C. Gastric acid stimulating properties D. Neuromuscular blocking properties

Answer: B

28. If an unknown plant extract increases charcoal transit from 60% to 92%, the extract possesses:

A. Anticholinergic properties B. Prokinetic / laxative / propulsive properties C. Opioid-like properties D. Sympathomimetic properties

Answer: B

29. Which parasympathomimetic (cholinomimetic) agent would dramatically accelerate charcoal transit in this assay?

A. Scopolamine (hyoscine) B. Neostigmine or bethanechol C. Epeprazole D. Propranolol

Answer: B

30. How does neostigmine accelerate gastrointestinal motility?

A. Reversibly inhibits acetylcholinesterase, preventing acetylcholine degradation and augmenting muscarinic transmission B. Directly blocks dopamine receptors in the brainstem C. Inactivates calcium channels D. Destroys the myenteric plexus

Answer: A

31. Castor oil accelerates intestinal motility primarily because its active metabolite, ricinoleic acid:

A. Irritates the intestinal mucosa and activates EP_3/EP_4 prostanoid receptors, stimulating powerful peristalsis B. Chemically binds all water in the lumen C. Directly stimulates central dopamine receptors D. Blocks the action of acetylcholine

Answer: A

32. Senna and bisacodyl belong to which class of laxatives?

A. Bulk-forming laxatives B. Stimulant / irritant laxatives C. Osmotic laxatives D. Stool softeners

Answer: B

33. Magnesium sulfate (Epsom salt) increases transit and causes diarrhea primarily by:

A. Stimulating beta-2 adrenergic receptors B. Osmotic water retention in the intestinal lumen, distending the bowel and stimulating peristalsis C. Inhibiting the release of motilin D. Direct cytotoxicity to enterocytes

Answer: B

34. Why must the excised small intestine NOT be forcefully stretched during measurement?

A. Stretching would destroy the charcoal dye chemically B. Forceful traction introduces artificial variability into the total length and transit measurements C. The intestine cannot stretch under any circumstances D. It causes immediate bacterial overgrowth

Answer: B

35. What is the anatomical landmark marking the start of the small intestine?

A. Gastroesophageal junction B. Gastroduodenal junction (pyloric sphincter) C. Ligament of Treitz D. Ileocecal valve

Answer: B

36. What is the anatomical landmark marking the end of the small intestine?

A. Pyloric sphincter B. Ileocecal junction (valve) C. Rectosigmoid junction D. Anal canal

Answer: B

37. A group of 5 mice treated with an unknown drug had transit percentages of 30%, 35%, 28%, 32%, and 35%. What is the mean % transit?

A. 28.0% B. 32.0% C. 35.0% D. 40.0%

Answer: B

38. Which gastrointestinal peptide hormone stimulates migrating motor complexes (MMCs) during the fasting state?

A. Secretin B. Motilin C. Cholecystokinin (CCK) D. Gastric inhibitory peptide (GIP)

Answer: B

39. In clinical pharmacology, which antispasmodic is commonly prescribed to relieve abdominal cramping in irritable bowel syndrome (IBS)?

A. Metoclopramide B. Hyoscine butylbromide (Buscopan) C. Domperidone D. Senna

Answer: B

40. The overall conclusion of Experiment 35 is that:

A. Gastrointestinal transit is fixed and cannot be modified by drugs B. The charcoal meal test is a reliable quantitative assay for evaluating drug-induced stimulation (prokinetics) or inhibition (antispasmodics, opioids) of intestinal peristalsis C. Mice have no cholinergic receptors in their gut D. Only oral drugs affect the gastrointestinal tract

Answer: B